Phase 1 Report — NoSQL ETL Framework

Phase 1 Report

Multi-Pipeline ETL & Reporting Framework for Web Server Log Analytics

Course: DAS 839 — NoSQL Systems, End-Semester Project **Phase:** 1 (architecture + working prototype) **Pipeline implemented in Phase 1:** MongoDB **Pipelines deferred to Phase 2:** Apache Pig, Apache Hive, Hadoop MapReduce

---

Executive Summary

We have built a runnable prototype of the multi-pipeline ETL framework with the **MongoDB** backend implemented end-to-end and the other three backends stubbed against a shared `Pipeline` interface. On the full NASA HTTP access logs (Jul95 + Aug95 = **3,461,613 lines**), the system parses, batches (`batch_size = 100,000` → 35 batches), executes the three mandated queries, and loads the results into PostgreSQL in **~114 seconds**. The framework's design choices (native ingestion + native batching + an engine-agnostic loader) are deliberate so that Phase 2 backends can plug in without altering parsing semantics, schema, or reporting code.

This report addresses the six discussion areas the project statement asks for in the Phase 1 review:

1. Overall design / architecture
2. Parsing strategy
3. ETL workflow
4. Batching approach
5. Relational reporting schema
6. Plan for ensuring equivalence across pipelines

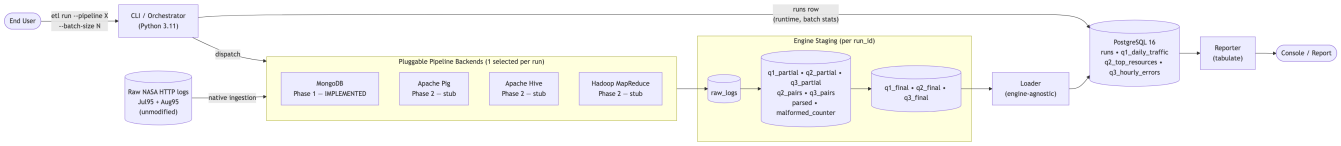
It closes with the working-prototype evidence from the demo run.

---

1. Overall Design

The system is a thin Python orchestrator around four pluggable execution backends. The orchestrator is responsible for I/O coordination, timing, and writing the run-metadata row to PostgreSQL; **all data processing happens inside the chosen backend** (MongoDB in Phase 1). Final results land in four PostgreSQL tables that are written by a backend-agnostic loader.

1.1 Component diagram



1.2 Module layout

```
NoSQL_Project_ETL_Analysis_Pipeline/
├── PARSING_SPEC.md              ← frozen rules; every backend must comply
├── PHASE1_REPORT.md             ← this document
├── docker-compose.yml           ← mongo:7 + postgres:16 (alternative to local installs)
├── pyproject.toml
├── sql/schema.sql               ← runs / q1 / q2 / q3 DDL
├── src/
│   ├── cli.py                   ← argparse: `run`, `report`
│   ├── orchestrator.py          ← dispatch + timer + runs row + cleanup
│   ├── loader.py                ← engine-agnostic q*_final → Postgres
│   ├── reporting/report.py      ← tabulate output
│   └── pipelines/
│       ├── base.py              ← Pipeline ABC + RunStats dataclass
│       ├── mongodb/             ← Phase 1: full implementation
│       │   ├── etl_pipeline.py  ← wires aggregations + ingestion
│       │   └── aggregations.py  ← pure-data agg builders
│       └── pig/, hive/, mr/     ← Phase 2: stubs raising NotImplementedError
└── tests/                       ← parsing + batch-id + e2e (13 passing)
```

The `Pipeline` ABC (`src/pipelines/base.py`) defines the five lifecycle methods every backend must implement (`ingest`, `aggregate`, `final_merge`, `collect_run_stats`, `cleanup`). The orchestrator and loader only ever talk to this interface, which is what makes the framework genuinely engine-pluggable.

1.3 Decisions & rationale

Decision	Choice	Reason
Orchestrator language	Python 3.11	Best `argparse` / `subprocess` story, native pymongo and psycopg2-binary
Result store	PostgreSQL 16	Project allows MySQL or Postgres; chose Postgres for `NUMERIC(20,6)` and `TIMESTAMPTZ`
Batching strategy	**Native, in-engine, logical** (computed by Mongo, not pre-sliced by Python)	Compliance with the project's "core data processing must genuinely happen in the chosen execution technology" rule
`batch_id` formula	`floor((ordinal - 1) / N) + 1`	Single integer formula, identical across all four engines
Distinct counts	**Exact**, via staged distinct-pair collections	Rejects HLL / probabilistic drift across engines
Phase 1 backend	MongoDB	Easiest path to a fully-runnable pipeline; aggregation framework natively supports regex parsing + windowed ranking

2. Parsing Strategy

Parsing is **fully specified in `PARSING_SPEC.md`** and frozen for the duration of the project. Every backend (Phase 1 Mongo and Phase 2 Pig / Hive / MR) must implement these rules byte-for-byte. This is what makes cross-engine equivalence achievable.

2.1 Master regex (POSIX-compatible)

```
^(\S+) \S+ \S+ \[([^\]]+)\] "([^"]*)" (\d{3}|-) (\d+|-)$
```

Capture groups: `host`, `timestamp_raw`, `request_raw`, `status_raw`, `bytes_raw`. The same regex string is used as `$regexFind` input in MongoDB, will be passed to `regexp_extract` in Hive, `REGEX_EXTRACT_ALL` in Pig, and `Pattern.compile` in MapReduce.

2.2 Field-level rules

Field	Source	Rule
`host`	capture 1	first whitespace-delimited token
`timestamp_raw`	capture 2	parsed via `%d/%b/%Y:%H:%M:%S %z`
`log_date`, `log_hour`	derived	extracted in the **timestamp's own offset** — no UTC conversion (so engines with different default TZ behaviour produce identical results)
`http_method` / `resource_path` / `protocol_version`	capture 3 split on whitespace	tokens 1, 2, 3 — missing tokens become `NULL`, **not malformed**
`bytes_transferred`	capture 5	`"-"` → `0`; numeric otherwise
`status_code`	capture 4	`"-"` → record is **malformed**

2.3 Malformed records

A record is malformed iff (a) the master regex fails to match, **or** (b) `status_raw == "-"`. Malformed records are **counted** (`malformed_count` in `runs`) and dropped from aggregation; they never reach Q1/Q2/Q3. They are **not** removed from the ordinal stream — that's important for batch-boundary stability (see §4).

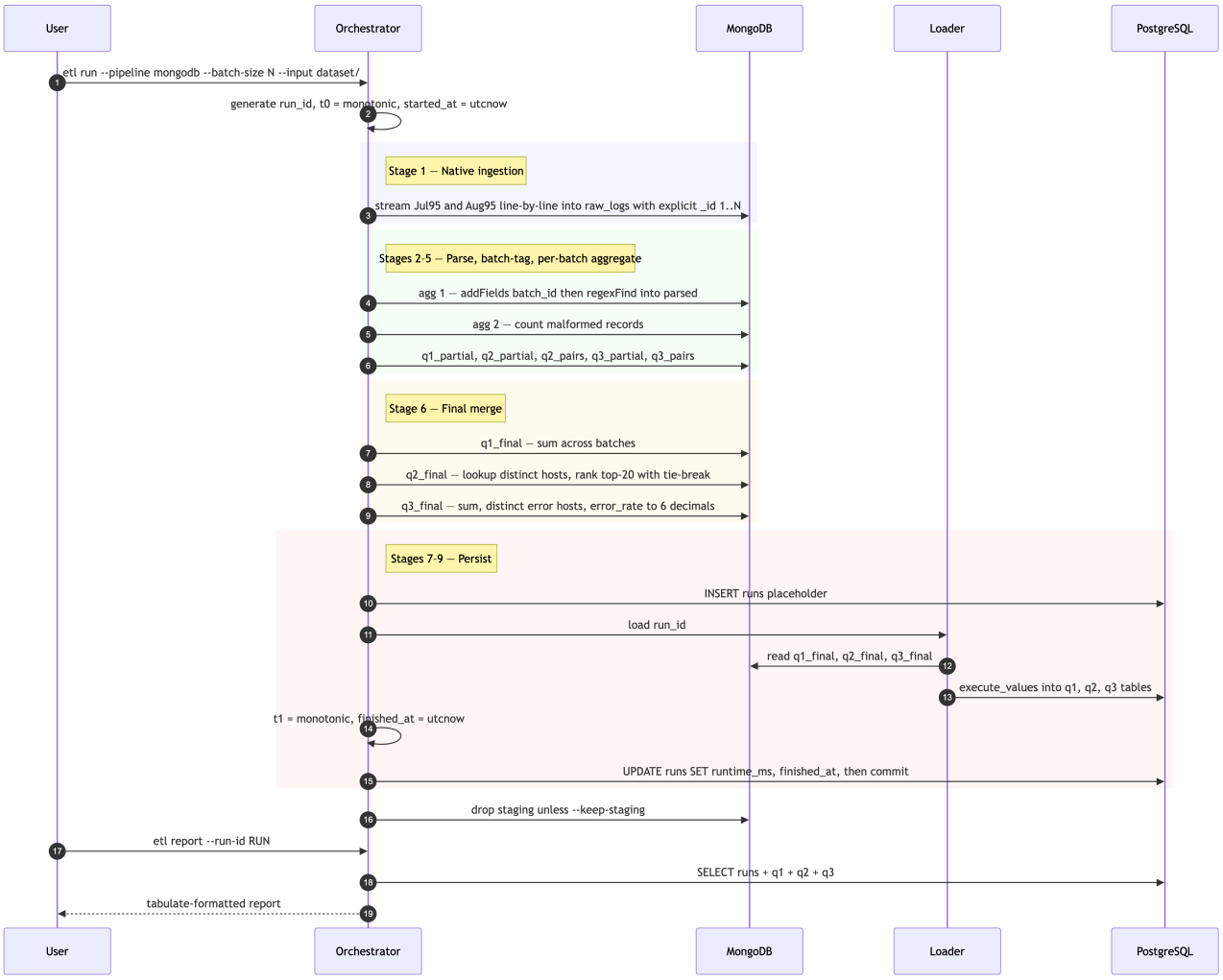
2.4 Validation evidence

`tests/test_parsing.py` runs the exact `MASTER_REGEX` constant from `aggregations.py` against 11 hand-crafted lines (5 valid + 2 with `bytes='-'` + 2 with `status='-'` + 2 truly malformed). It asserts `malformed_count == 4` and verifies capture-group correctness, including that short-request lines are NOT malformed. On the real dataset, the run reported `malformed_count = 33` out of 3,461,613 lines (≈ 0.001%), consistent with the known small fraction of bad lines in the NASA traces.

3. ETL Workflow

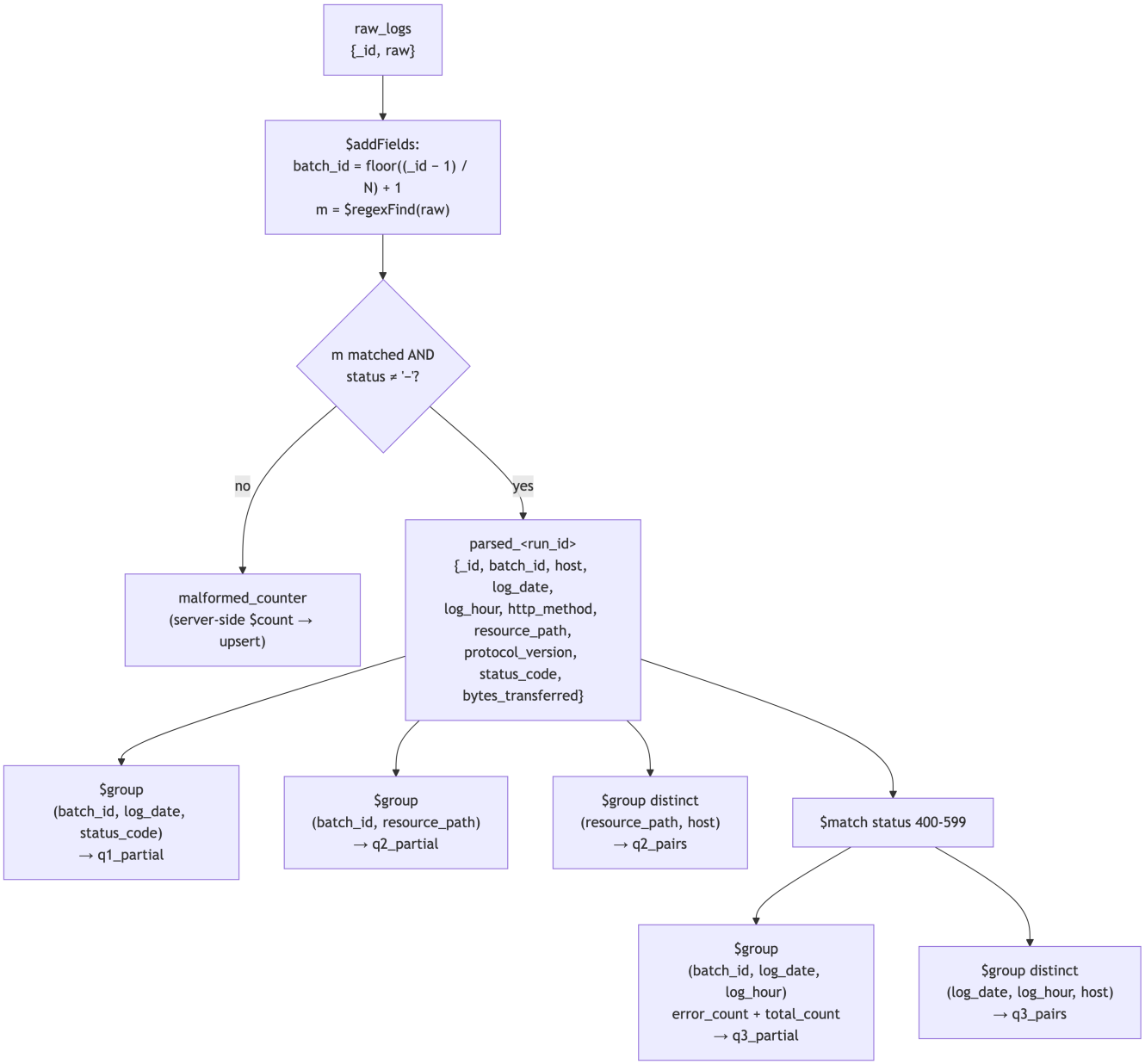
The workflow follows ten stages, of which stages 2–7 happen entirely inside MongoDB. Python only orchestrates and never touches a parsed field.

3.1 Sequence diagram



3.2 Per-batch dataflow inside MongoDB

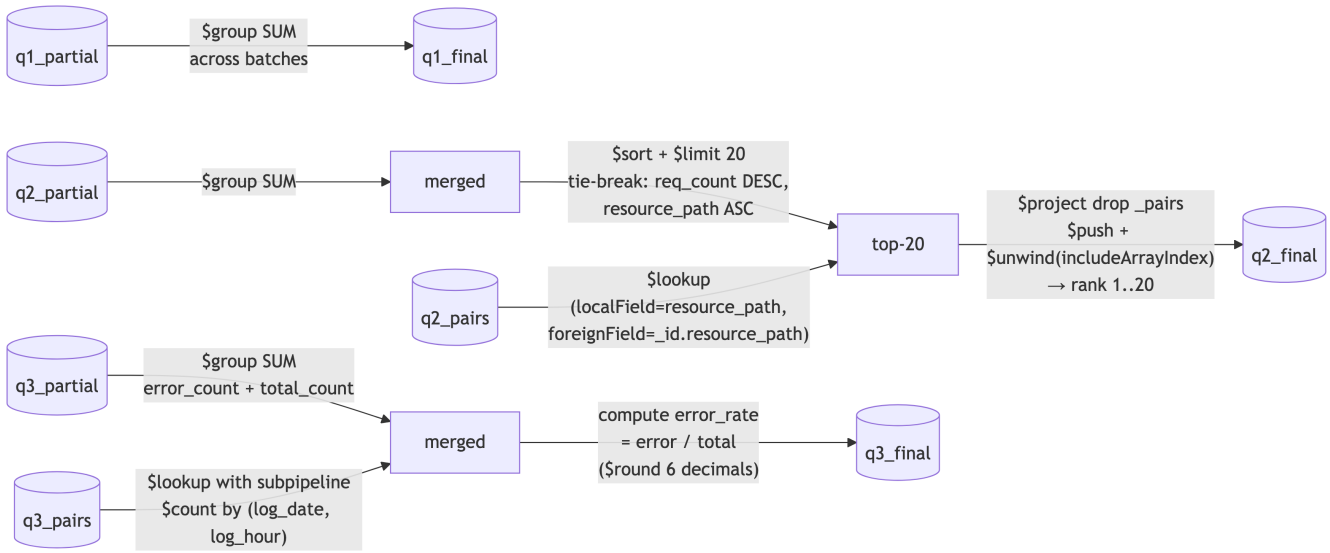
Inside the `aggregate()` step, after parse-and-tag has written `parsed_<run_id>`, five independent aggregations build the per-batch staging:



Two design notes worth highlighting:

- `q3_partial` carries **both** `error_request_count` (conditional sum: 1 iff 400 ≤ status ≤ 599) and `total_request_count` (unconditional sum). This avoids a second pass over `parsed` to compute the denominator of `error_rate`.
- `q2_pairs` and `q3_pairs` are **global** distinct collections (one row per unique pair / triple). They are not per-batch. The `$group` step itself dedupes across all input documents.

3.3 Final merge



Why `$push` + `$unwind` and not `$setWindowFields($documentNumber)` for Q2 ranking? `$documentNumber` requires a *single-element* `sortBy`, which can't express our two-key tie-break (`request_count DESC, resource_path ASC`). The `$sort + $limit(20)` step already establishes the canonical order; `$push` preserves that order, and `$unwind` with `includeArrayIndex` exposes the position as the rank. We also `$project: {_pairs: 0}` between `$lookup` and `$push` to discard the per-resource host arrays — without that, hot resources with 80k+ distinct hosts would push the in-memory accumulator past the 100 MB limit.

---

4. Batching Approach

4.1 Definition

The project statement defines batch size as **the number of input log records processed in one batch** (not bytes, not files, not aggregation chunks). Batch IDs start at 1, increase sequentially, and the last batch may be partial but is still counted.

4.2 Native, in-engine, logical batching

Two important properties of our implementation:

- **Native** — Python does not pre-slice the input. It only streams raw text into one collection (`raw_logs_<run_id>`) with explicit monotonically-increasing `_id` values 1..N. The batch boundaries are derived **inside MongoDB** via `$addFields` (see formula below).
- **Logical** — there is no physical separation: parsed records all sit in `parsed_<run_id>`; their batch is just a tag (`batch_id`). The same data can be re-batched at a different `N` simply by re-running the aggregation with a different parameter, no re-ingestion needed.

This satisfies the implementation constraint that "*core data processing must genuinely happen using the selected execution technology*". Pre-tokenising or pre-slicing in Python would be partial substitution, which the project statement disallows.

4.3 The formula

```
ordinal  = 1-indexed position of the record across the input files (lex-sorted)
batch_id = floor((ordinal - 1) / N) + 1
```

Worked example for `N = 10`, ordinals 1..25:

```
ordinal  : 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25
batch_id : 1  1  1  1  1  1  1  1  1  1  2  2  2  2  2  2  2  2  2  2  3  3  3  3  3
```

Batches 1 and 2 are full (10 records each); batch 3 is partial (5 records). All three count toward `num_batches`. (Verified by `tests/test_batch_id.py`.)

4.4 Why ordinals are computed across all input lines

Including malformed lines. If we computed ordinals only over parsed records, then any change in the number of malformed lines (e.g. fixing a regex) would shift batch boundaries non-deterministically. By reserving an ordinal for every input line (malformed or not) and only excluding malformed records from aggregation later, batch boundaries are stable and reproducible.

4.5 Per-engine ordinal primitive (forward-looking)

Engine	Ordinal source
MongoDB (Phase 1)	explicit `_id = 1..N` set during ingestion
Hive	`ROW_NUMBER() OVER (ORDER BY <stable_input_order>)`
Pig	`RANK <relation> BY ... DENSE` over the same stable order
MapReduce	line-offset emitted by the input format, then a single-reducer 1..N assignment

The formula `floor((ordinal - 1) / N) + 1` is identical in every engine.

4.6 Reporting metadata

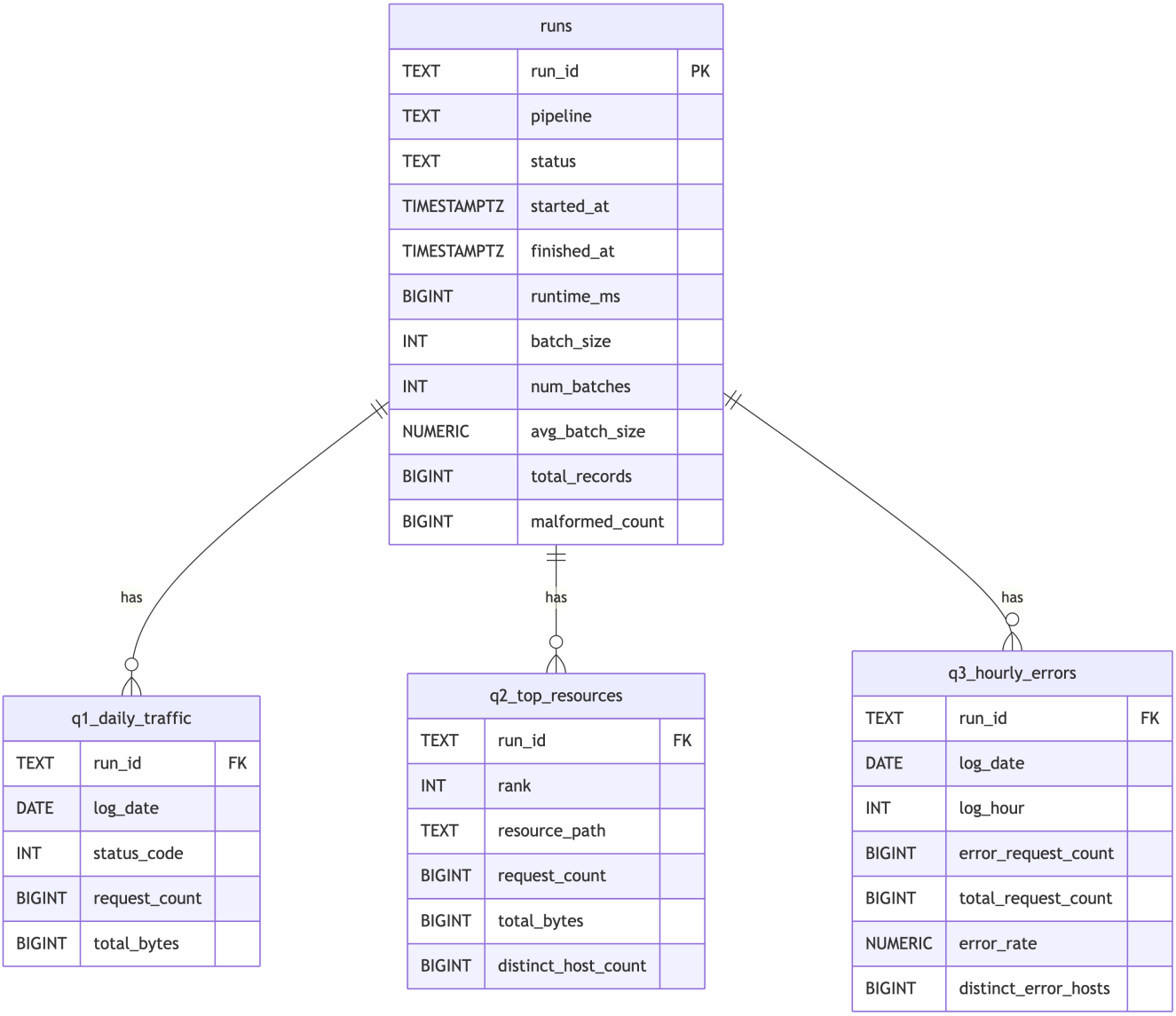
Field	Value (this run, N = 100,000)
`batch_size`	100,000
`num_batches`	35 — `max(batch_id)` over `parsed_<run_id>`
`avg_batch_size`	98,902.29 — `total_records / num_batches` per project statement
`total_records`	3,461,580 — parsed (excludes 33 malformed)
`malformed_count`	33

---

5. Reporting Database Schema

PostgreSQL 16. Four tables: one for run metadata, three for query outputs. Cascaded foreign keys clean up child rows when a `runs` row is deleted.

5.1 ER diagram



5.2 Schema verbatim

The full DDL is in `sql/schema.sql`. Key points:

- `runs.run_id` is `TEXT` (we use `uuid4().hex`); every query result table references it.
- `runs.status` is `'running'` while a run is in progress and updated to `'completed'` only after the loader writes Q1/Q2/Q3 successfully. If a run crashes mid-flight, the row stays as `'running'` — distinguishable from a finished result.
- `q1_daily_traffic` PK is `(run_id, log_date, status_code)` — guarantees one row per (run, day, status).
- `q2_top_resources` PK is `(run_id, rank)` — exactly 20 rows per run.

- `q3_hourly_errors.error_rate` is `NUMERIC(20, 6)` — locks 6-decimal precision per the parsing spec.
- All FKs include `ON DELETE CASCADE` so dropping a `runs` row cleans up its children.

5.3 Project-statement requirements satisfied

The project statement requires the relational store to record **pipeline name, run identifier, batch identifier, time of execution**. Mapping:

Required	Where in our schema
pipeline name	`runs.pipeline`
run identifier	`runs.run_id` (referenced by all q* tables)
batch identifier	`runs.num_batches` + `runs.batch_size` (the formula recovers `batch_id` for any record)
time of execution	`runs.started_at`, `runs.finished_at`, `runs.runtime_ms`

Reporting metadata required (pipeline, query name, runtime, batch ID, batch size, average batch size) is all materialised in the `runs` row and shown in the report header.

---

6. Equivalence Plan Across Pipelines

The project's central requirement is that all four pipelines produce **the same Q1/Q2/Q3 outputs from the same input**. Our framework enforces this through five mechanisms:

1. **One frozen parsing spec.** `PARSING_SPEC.md` is the single source of truth — same regex, same timestamp rule (no UTC conversion), same `bytes='-' → 0` mapping, same malformed definition. Phase 2 backends will reuse the literal regex string.
2. **One batch-id formula.** `floor((ord - 1) / N) + 1` — a single integer expression every engine can implement using its native ordinal primitive (§4.5).
3. **Exact distinct counts.** Every engine must build a `(key, host)` distinct-pair stage and count exactly. No HLL, no `approx_distinct`. This is a frequent source of cross-engine drift; we cut it off at the design level.
4. **A shared `Pipeline` ABC.** `src/pipelines/base.py` defines five lifecycle methods. The orchestrator and loader only use this interface, so any engine that conforms is interchangeable. Phase 2 backends are stubs of this same ABC today.
5. **A common output contract.** Every backend must populate four well-known per-run collections / dirs: `q1_final_<run_id>`, `q2_final_<run_id>`, `q3_final_<run_id>`, plus the `RunStats` returned by `collect_run_stats`. The loader is engine-agnostic — it reads from these four locations regardless of which backend ran.

6.1 Phase 2 verification plan

Once Pig / Hive / MR backends are implemented, we will run all four against the **same `--batch-size` and same input**, then diff Q1/Q2/Q3:

```
-- pseudocode for the cross-engine diff (Phase 2)
SELECT log_date, status_code, request_count, total_bytes
```

```sql
FROM q1_daily_traffic WHERE run_id = '<mongo_run>'
EXCEPT
SELECT log_date, status_code, request_count, total_bytes
FROM q1_daily_traffic WHERE run_id = '<hive_run>';
-- expected: zero rows
```

Any non-empty diff exposes a divergence and must be traced back to a parsing-spec violation in the offending engine.

---

7. Phase 1 Demonstration

7.1 Run command

```bash
psql "postgresql://etl:etl@localhost:5432/etl" -f sql/schema.sql
python -m src.cli run --pipeline mongodb \
    --input ../dataset/ --batch-size 100000
python -m src.cli report --run-id <printed_id>
```

7.2 Run metadata (real output)

```
| field           | value                              |
|-----------------|------------------------------------|
| run_id          | 3a5e337e2d6f466590fda039afc86058 |
| pipeline        | mongodb                            |
| status          | completed                          |
| started_at      | 2026-04-28 04:55:30.218668+05:30 |
| finished_at     | 2026-04-28 04:56:59.657279+05:30 |
| runtime_ms      | 89438                              |
| batch_size      | 100000                             |
| num_batches     | 35                                 |
| avg_batch_size  | 98902.29                           |
| total_records   | 3461580                            |
| malformed_count | 33                                 |
```

7.3 Q1 sample (295 rows total)

log_date	status_code	request_count	total_bytes
1995-07-01	200	58 033	1 617 409 574
1995-07-01	302	2 568	218 465
1995-07-01	304	3 797	0
1995-07-01	404	316	0
1995-07-03	500	53	0
1995-08-31	200	…	…

7.4 Q2 — top 20 requested resources

rank	resource_path	request_count	total_bytes	distinct_host_count
1	/images/NASA-logosmall.gif	208 798	131 778 402	81 518
2	/images/KSC-logosmall.gif	164 976	169 364 272	85 161
3	/images/MOSAIC-logosmall.gif	127 916	40 594 653	54 904
4	/images/USA-logosmall.gif	127 082	25 993 422	54 498
5	/images/WORLD-logosmall.gif	125 933	73 796 052	54 054
6	/images/ksclogo-medium.gif	121 580	629 750 296	49 491
7	/ksc.html	83 918	563 742 661	29 191
8	/images/launch-logo.gif	76 009	114 690 489	42 558
…	…	…	…	…
20	/icons/blank.xbm	28 852	13 214 149	17 846

Top resource is `/images/NASA-logosmall.gif` — consistent with the NASA Kennedy Space Center site's structure (a logo image embedded on every page).

7.5 Q3 sample (1,350 rows total)

log_date	log_hour	error_request_count	total_request_count	error_rate	distinct_error_hosts
1995-07-01	0	24	3 565	0.006732	12
1995-07-01	1	…	…	…	…

7.6 Test suite

```
$ pytest -q
13 passed in 0.77s
```

- `tests/test_parsing.py` — 4 tests; verifies the master regex against the malformed-rule edge cases.
- `tests/test_batch_id.py` — 5 tests; verifies the batch-id formula across N = 1, 10, 1000, equal-to-total, and partial-last-batch cases.
- `tests/test_aggregations.py` — 4 tests; full ingest → aggregate → final-merge against a 15-line hand-crafted dataset (13 valid + 2 malformed). Asserts Q1 group counts, Q2 tie-break ordering (`/b` ranks before `/c` when both have `request_count = 3`), Q3 `error_rate` to 6 decimals, and run-stats invariants (`num_batches = 2` , `avg_batch_size = 6.5`).

---

8. Status & Phase 2 Roadmap

8.1 What's done in Phase 1

- ✅ Frozen parsing spec
- ✅ Postgres schema (4 tables, FKs, cascading delete, indexes, run-status column)
- ✅ `Pipeline` ABC + `RunStats` dataclass
- ✅ MongoDB backend: ingestion, parse-and-tag, per-batch partials, distinct-pair staging, final merge with ranked Q2 and 6-decimal error rate
- ✅ Engine-agnostic loader (Mongo `q*_final` → Postgres)
- ✅ Orchestrator with monotonic timing + transactional runs-row write + cleanup
- ✅ CLI with `run` and `report` subcommands
- ✅ Reporter (tabulate, GitHub flavour)
- ✅ 13-test smoke suite (parsing, batch-id, e2e)
- ✅ End-to-end run on real 3.46M-line dataset; all three result tables populated
- ✅ Code-review pass: premature-commit bug, connection-leak bug, whitespace-only request handling, idempotent ingestion, run-status tracking — all addressed (see §8.3)

8.3 Hardening applied after the in-tree code review

#	Issue	Fix
Loader committed too early	Loader called `pg_conn.commit()` before the runs-row UPDATE, so a subsequent failure left `runtime_ms = 0` permanently	Removed the commit; orchestrator owns the single-transaction commit
Connection leak	`psycopg2.connect()` as a context manager only manages the transaction, not the connection itself	Explicit `pg_conn.close()` in `finally` blocks (orchestrator + reporter)
Whitespace-only request line	`"  "` would set `http_method=""` instead of NULL	`req_tokens` now uses `$filter` to drop empty tokens before length check
Re-run safety	`insert_many(ordered=True)` aborted the whole batch on a duplicate `_id` retry	`ordered=False` so partial duplicates are reported but ingestion continues
Run-status tracking	A failed run left `runtime_ms = 0` indistinguishable from a real result	New `runs.status` column: `'running'` on placeholder INSERT, `'completed'` on UPDATE
Empty-input guard	`total_records == 0` would write a meaningless runs row	Orchestrator now raises `RuntimeError` before any DB write
Final-collection cleanup	`q*_final` Mongo collections lingered after load	Added to `_STAGING_PREFIXES`

8.2 Phase 2 work

Item	Owner	Notes
Apache Pig backend	TBD	

Item	Owner	Notes
		Same ABC; ordinal via `RANK ... DENSE`; same regex via `REGEX_EXTRACT_ALL`; same distinct-pair pattern via `DISTINCT` then `COUNT_STAR`
Apache Hive backend	TBD	Ordinal via `ROW_NUMBER()`; `regexp_extract` for parsing; native `COUNT(DISTINCT ...)` for Q2/Q3 distincts
Hadoop MapReduce backend	TBD	Two-pass: first pass assigns ordinals via single reducer; second pass parses and emits per-batch partials; combiners for Q1/Q2/Q3 sums
Cross-engine equivalence diff	TBD	SQL `EXCEPT` between any two `run_id`s; Q1/Q2/Q3 expected to be byte-identical
Comparative report (Phase 2 PDF)	TBD	Runtime / lines-of-code / difficulty per engine; observations on suitability
Video demo	TBD	Live demo of pipeline switching, with all four backends populating the same Postgres tables

---

9. Reproducing the Phase 1 Demo

```
# 1. Mongo (local install via brew, or `docker compose up -d mongo`)
brew services start mongodb-community@7.0

# 2. Postgres (local brew install, or compose-up postgres)
psql -d postgres -c "CREATE ROLE etl LOGIN PASSWORD 'etl';"
psql -d postgres -c "CREATE DATABASE etl OWNER etl;"
psql -U etl -d etl -h localhost -f sql/schema.sql

# 3. Python
python3.11 -m venv .venv && source .venv/bin/activate
pip install -e ".[dev]"

# 4. Tests
pytest -q

# 5. Real demo
python -m src.cli run --pipeline mongodb \
    --input ../dataset/ --batch-size 100000
python -m src.cli report --run-id <printed_id>
```

The full setup is also documented in `README.md`.